THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – V : Refrigeration & Air Conditioning

★ Note: Syllabus: Introduction to refrigeration: air & vapour-compression systems, components, COP, refrigerants. Air conditioning: psychrometric properties & processes, sensible & latent heat loads, SHF, human comfort, effective temperature, comfort chart, load calculations

1. Introduction to Refrigeration

Refrigeration is the process of removing heat from a space or substance to bring it to a temperature lower than its surroundings and to maintain it there. It needs a work (or heat) input and a working fluid called the refrigerant.

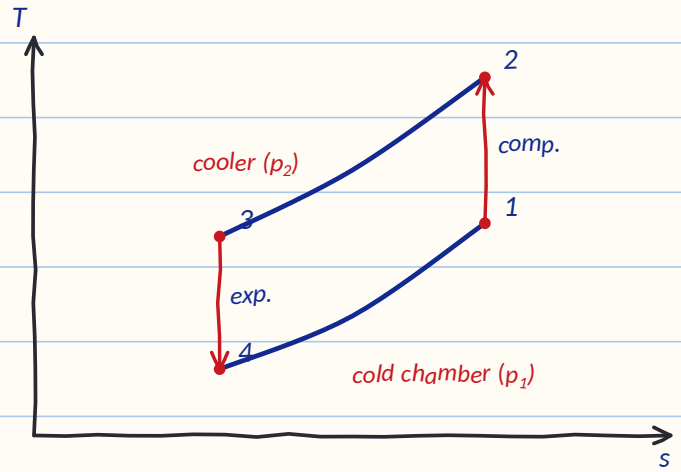
- Applications: food preservation, cold storage, ice plants, air conditioning, chemical industries, medical (vaccines, blood banks), cryogenics.
- Refrigerating effect (RE): heat extracted from the cold space per kg of refrigerant, kJ/kg. Capacity: kW or ton of refrigeration (1 TR = 3.517 kW = 211 kJ/min).

$$\text{COP} = \text{Refrigerating effect} / \text{Work input} = Q_2/W$$

- Reversed Carnot cycle (ideal limit): $\text{COP}_{max} = T_2/(T_1 - T_2)$, T_2 = evaporator temperature, T_1 = condenser temperature (K).
- Methods: vapour compression, vapour absorption, air refrigeration (gas cycle), thermoelectric, steam-jet.

2. Air Refrigeration – Bell–Coleman Cycle

Air is the working fluid (no phase change). It is the reversed Brayton (Joule) cycle, used in aircraft cooling and in cryogenic plants.



Air refrigeration (Bell–Coleman)

Reversed Brayton cycle

$RE = c_p(T_1 - T_4)$

$W = c_p[(T_2-T_1)-(T_3-T_4)]$

$COP = T_4 / (T_3 - T_4)$

$= 1/(r_p^{(\gamma-1)/\gamma} - 1)$

→ 1-2 Isentropic compression in compressor: $P_1 \rightarrow P_2$, $T_1 \rightarrow T_2$.

→ 2-3 Constant-pressure cooling in the cooler (heat Q_1 rejected to surroundings).

→ 3-4 Isentropic expansion in the expander: T falls to T_4 (lowest temperature).

→ 4-1 Constant-pressure heating in the cold chamber (refrigerating effect, Q_2 absorbed).

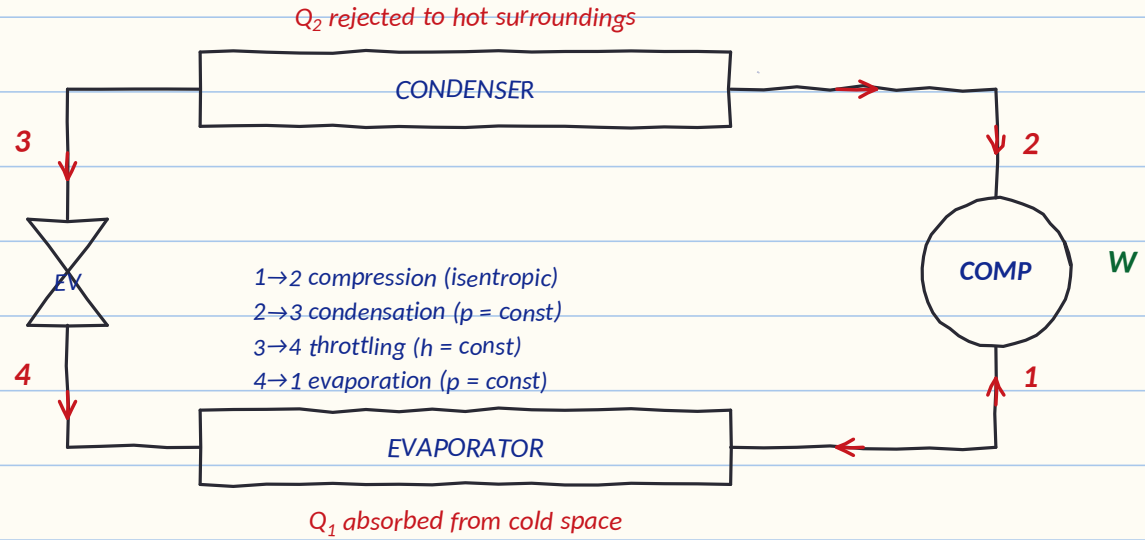
$$RE = c_p(T_1 - T_4) \qquad W = c_p[(T_2 - T_1) - (T_3 - T_4)]$$

$$COP = (T_1 - T_4) / [(T_2 - T_1) - (T_3 - T_4)] = 1/(r_p^{(\gamma-1)/\gamma} - 1)$$

- Advantages: air is cheap, non-toxic, non-flammable; light in weight – used on aircraft (cabin pressurised air is already available).
- Disadvantages: very low COP (< 1 for high r_p); large volume of air required; sensible-heat only, no latent heat; bulky equipment.
- Since the heat exchange is at varying temperature (not isothermal), COP is always lower than the Carnot COP between the same extremes.

3. Vapour Compression Refrigeration (VCR) System

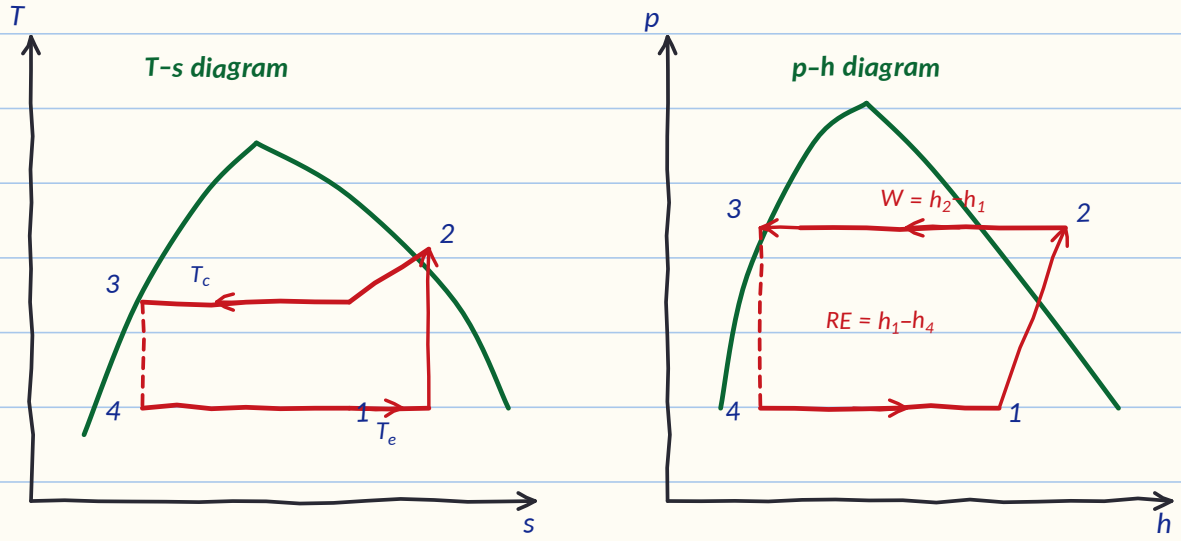
The most widely used system – the refrigerant undergoes phase change (evaporates, condenses), using the large latent heat.



Components

Component	Function
Compressor	Raises P and T of vapour (reciprocating, rotary, screw, centrifugal, scroll)
Condenser	Rejects heat; vapour → liquid at high P (air-/water-cooled)
Expansion valve	Throttles liquid to low P; T falls (capillary tube, TXV)
Evaporator	Liquid absorbs heat from the cold space and evaporates

Processes (T-s and p-h diagrams)



→ 1-2 Isentropic compression of dry saturated vapour to superheated vapour ($W = h_2 - h_1$).

→ 2-3 Constant-pressure cooling and condensation to saturated liquid in the condenser ($Q_1 = h_2 - h_3$).

→ 3-4 Throttling (isenthalpic): $h_3 = h_4$, irreversible; part of the liquid flashes to vapour.

→ 4-1 Constant-pressure, constant-temperature evaporation ($RE = h_1 - h_4$).

$$RE = h_1 - h_4 = h_1 - h_3 \qquad W = h_2 - h_1$$

$$COP = (h_1 - h_4) / (h_2 - h_1) \qquad Q_{cond} = h_2 - h_3 = RE + W$$

- Why a throttle valve instead of an expander? The work recovered from a small liquid-vapour expansion is very small, and the valve is simple, cheap and controllable.
- Dry compression (vapour enters dry) is preferred to wet compression (liquid droplets can damage the compressor).
- Sub-cooling the liquid before the valve and superheating the vapour before compression increase RE; superheat adds some work.
- Vapour compression advantages over air refrigeration: high COP,

compact, uses latent heat, nearly constant evaporator temperature.

Effect of operating conditions

- Lower evaporator pressure ⇒ lower RE and more work per kg ⇒ COP decreases. Higher condenser pressure ⇒ more work, less RE ⇒ COP decreases.
- Practical (actual) cycle deviates: pressure drops in pipes, heat gains, non-isentropic compression ($\eta_{isen} < 1$), sub-cooling and superheating.

4. Refrigerants

A refrigerant is the working fluid that absorbs heat at low T in the evaporator and rejects it at high T in the condenser.

Designation (ASHRAE R-number)

- R-xyz: x = C − 1, y = H + 1, z = F. R-12 = CCl_2F_2, R-22 = $CHClF_2$, R-134a = $CH_2F–CF_3$ (a = isomer).
- Inorganic: 700 + molecular mass: R-717 = ammonia (NH_3), R-718 = water, R-744 = CO_2. Hydrocarbons: R-290 propane, R-600a isobutane.

Desirable properties

- Thermodynamic: high latent heat, low freezing point, high critical temperature, moderate pressures (above atmosphere in evaporator), low specific volume of vapour, high COP.
- Physical / chemical: non-toxic, non-flammable, non-corrosive, chemically stable, miscible with lubricating oil, high thermal conductivity, easy leak detection, low cost.
- Environmental: zero ODP (ozone depletion potential) and low GWP (global warming potential).

Refrigerant	Type	Remarks
R-12 (CFC)	synthetic	high ODP – banned (Montreal Protocol)
R-22 (HCFC)	synthetic	low ODP, being phased out
R-134a (HFC)	synthetic	ODP 0, GWP ≈ 1430; domestic refrigerators
R-717 (NH_3)	natural	excellent COP, toxic; industrial
R-744 (CO_2)	natural	non-toxic, very high pressure
R-290 / R-600a	natural	flammable, low GWP
R-1234yf (HFO)	synthetic	GWP < 1; automotive A/C

- *Secondary refrigerants* (brine – NaCl, $CaCl_2$; glycol; chilled water) carry cold from the evaporator to the load; they do not change phase.

5. Introduction to Air Conditioning

Air conditioning is the simultaneous control of air temperature, humidity, cleanliness and motion in a space to suit the occupants or a process.

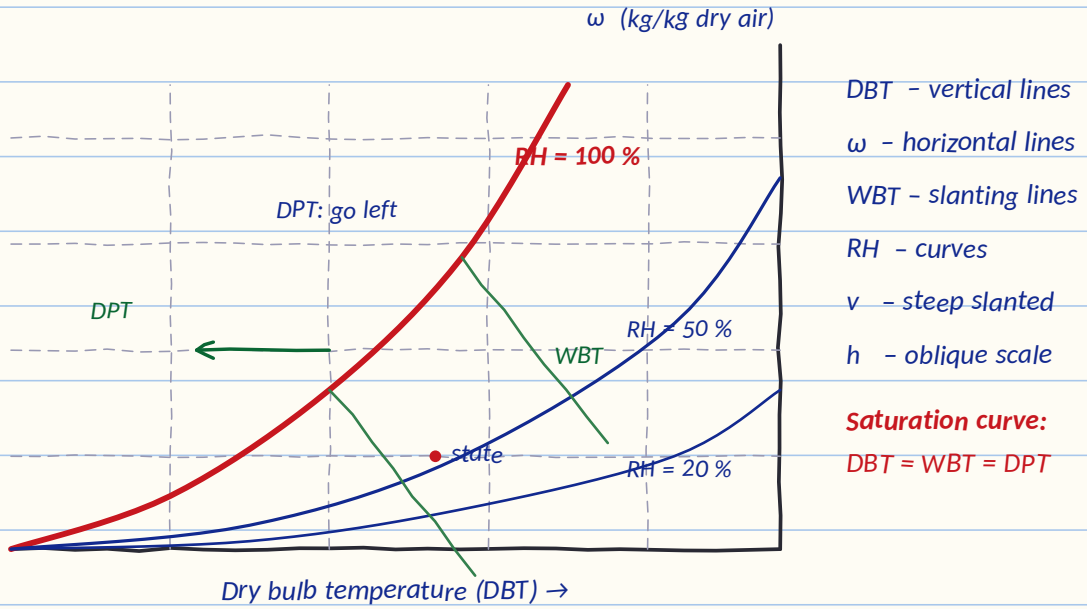
- Comfort air conditioning – for human comfort (homes, offices, theatres). Industrial A/C – for process requirements (textile, printing, pharma, electronics).
- By season: *summer* (cooling + dehumidification), *winter* (heating + humidification), *year-round*.
- Equipment: filter, cooling coil, heating coil, humidifier, fan / blower, ducts, diffusers, thermostat, refrigeration plant.

6. Psychrometric Properties

Moist air = dry air + water vapour. By Dalton's law the total pressure $P = p_a + p_v$.

Property	Definition / formula
Dry bulb temp (DBT) t	temperature shown by ordinary thermometer
Wet bulb temp (WBT) t_w	thermometer with wetted wick; evaporative cooling
Dew point temp (DPT)	temperature at which vapour starts condensing at constant P
Relative humidity φ	p_v/p_g at the DBT (p_g = saturation pressure)
Humidity ratio ω	$m_v/m_a = 0.622\ p_v/(P - p_v)$ kg/kg dry air
Degree of saturation μ	ω/ω_s at same DBT
Specific enthalpy h	$1.005\ t + \omega(2500 + 1.88\ t)$ kJ/kg dry air
Specific volume v	m^3 per kg dry air = R_aT/p_a

- DBT ≥ WBT ≥ DPT; for saturated air all three are equal and φ = 100 %.
- Carrier's equation (from a sling psychrometer): $p_v = p_w - (P - p_w)(t - t_w)/(1544 - 1.44\ t_w)$.
- Humid specific heat $c_{pm} = 1.005 + 1.88\ \omega \approx 1.022$ kJ/kg·K.



- The psychrometric chart shows these properties together: given any two independent properties, the rest are read from the chart.

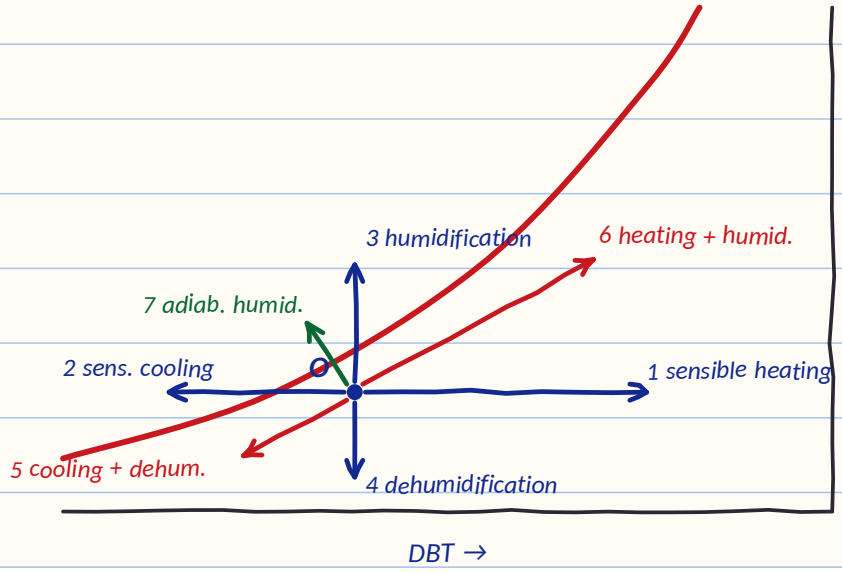
Example

$P = 1.01325$ bar, $p_v = 0.02$ bar, $t = 30$ °C ($p_g = 0.04246$ bar)

$\omega = 0.622 \times 0.02/(1.01325 - 0.02) = 0.0125$ kg/kg ; $\varphi = 0.02/0.04246 = 47$ %

$h = 1.005 \times 30 + 0.0125(2500 + 1.88 \times 30) = 62.1$ kJ/kg dry air

7. Psychrometric Processes



Sensible heat:

$Q_s = m\, c_{pm}(t_2 - t_1)$

Latent heat:

$Q_l = m\, h_{fg}(\omega_1 - \omega_2)$

SHF = $Q_s/(Q_s+Q_l)$

RSHF, GSHF, ESHF

Process	Path on chart	Example
Sensible heating	horizontal →, ω const	heating coil
Sensible cooling	horizontal ←, ω const (t > DPT)	dry cooling coil
Humidification	vertical ↑, t const	steam injection
Dehumidification	vertical ↓, t const	chemical absorbent
Cooling + dehumid.	down-left ↙	summer A/C coil
Heating + humid.	up-right ↗	winter A/C
Adiabatic humid.	along WBT line ↖	air washer

Adiabatic mixing of two air streams

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_2) h_3$$

$$\dot{m}_1 \omega_1 + \dot{m}_2 \omega_2 = (\dot{m}_1 + \dot{m}_2) \omega_3$$

$$t_3 \approx (\dot{m}_1 t_1 + \dot{m}_2 t_2)/(\dot{m}_1 + \dot{m}_2)$$

On the chart the mixed state lies on the straight line joining the two states, dividing it in the inverse ratio of the masses.

2 kg/s at 30 °C, ω = 0.020 with 1 kg/s at 20 °C, ω = 0.008:

$t_3 = (60 + 20)/3 = 26.7$ °C ; $\omega_3 = (0.040 + 0.008)/3 = 0.016$

Cooling coil – bypass factor

$$BF = (t_3 - t_{ADP})/(t_1 - t_{ADP}) \qquad \text{Contact factor} = 1 - BF$$

- ADP (apparatus dew point) = effective surface temperature of the coil. Smaller BF ⇒ better coil (more rows, lower air velocity).

8. Sensible and Latent Heat Loads – SHF

- Sensible heat changes DBT without changing moisture. Latent heat is associated with change in humidity ratio (moisture addition/removal).

$$Q_s = \dot{m}_a c_{pm} \Delta t = 1.23 \dot{V} \Delta t \quad (kW, \dot{V} \text{ in } m^3/s)$$

$$Q_l = \dot{m}_a \Delta\omega h_{fg} \approx 3000 \dot{V} \Delta\omega \quad (kW)$$

★ Note: Sensible Heat Factor SHF = $Q_s/(Q_s + Q_l) = Q_s/Q_{total}$. A low SHF means a large latent load (humid climate, crowded halls).

- RSHF = room SHF; GSHF = grand SHF including fresh-air load on the coil; ESHF = effective SHF taking coil bypass into account.
- SHF line on the chart: drawn from the room state with the slope given by the protractor; it fixes the supply air state and the ADP.

Example: $\dot{V} = 2\ m^3/s$, $\Delta t = 10\ K$, $\Delta\omega = 0.0015$

$Q_s = 1.23 \times 2 \times 10 = 24.6\ kW$; $Q_l = 3000 \times 2 \times 0.0015 = 9\ kW$

$SHF = 24.6/33.6 = 0.73$

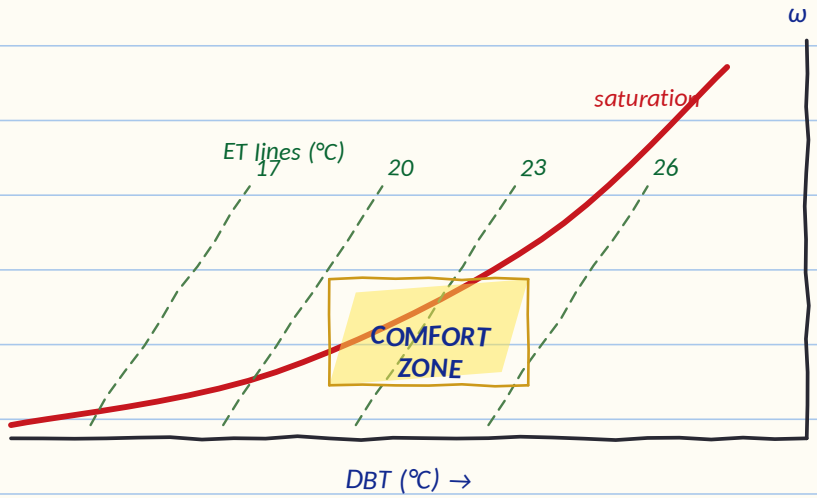
9. Human Comfort

The human body produces about 100 W of heat at rest (metabolism) and must lose it to keep the core near 37 °C. Heat loss: convection, radiation, evaporation (skin and breath), small conduction.

- Comfort is the condition of mind that expresses satisfaction with the thermal environment (ASHRAE).
- Factors: dry bulb temperature, humidity, air velocity, mean radiant temperature, activity level, clothing, age, sex, and cleanliness (odour, dust, CO_2).
- High humidity restricts sweat evaporation (hot-humid feels worse); low humidity dries the skin; air motion raises evaporation and convection.
- Recommended: DBT 22–27 °C, RH 40–60 %, air velocity 5–8 m/min, fresh air ≈ 7–10 L/s per person.

10. Effective Temperature and Comfort Chart

★ Note: Effective Temperature (ET) is an empirical index combining DBT, humidity and air velocity: it is the temperature of still, saturated air that gives the same sensation of warmth or cold.



Effective temperature (ET):
index combining DBT, humidity & air velocity → same feel

Typical comfort zone:
DBT 22–27 °C, RH 30–70 %
Air velocity ≈ 5–8 m/min

- Developed by Houghten and Yaglou (ASHVE). The comfort chart has DBT on the abscissa, WBT on the oblique scale, with ET lines and a shaded comfort zone.
- Persons at the same ET feel the same, though DBT and RH differ. Comfort zone ET ≈ 21–25 °C (approx.); winter zone is slightly lower.
- Limitations: overestimates humidity effect at low temperatures; ignores radiation – replaced by the corrected effective temperature (globe temperature) and PMV-PPD indices.

11. Comfort Air Conditioning and Load Calculations

- Summer: air is cooled and dehumidified by a cooling coil (process ↙), then reheated slightly if required, and supplied to the room.
- Winter: air is preheated, humidified (steam / spray) and heated to the supply state (process ↗).

Cooling load components

Sensible loads	Latent loads
Heat conduction through walls, roof, glass	Moisture from occupants (sweat, breath)
Solar radiation through windows	Moisture in infiltration & ventilation air
People (≈ 70–75 W sensible each)	Cooking, steam, open water surfaces
Lights, motors, appliances	Moisture from equipment / process
Infiltration and ventilation air	People (≈ 45–55 W latent each)

- Steps: (1) fix inside & outside design conditions; (2) compute envelope gains $Q = U\,A\,\Delta T_{eq}$; (3) add internal gains; (4) add ventilation & infiltration loads; (5) add safety factor and duct gains (5–10 %); (6) find SHF, supply air quantity, coil capacity (TR).

$$\text{Supply air: } \dot{V} = Q_{s,room} / [1.23\,(t_{room} - t_{supply})] \quad (m^3/s)$$

- Winter heating load is mainly conduction loss $Q = U\,A\,(t_i - t_o)$ plus ventilation/infiltration load.

Solved load example

Office, 20 persons: people $20 \times 75 = 1.5$ kW sensible, $20 \times 55 = 1.1$ kW latent

Lights 2 kW, equipment 1.5 kW, solar + transmission 6 kW

Sensible = 1.5 + 2 + 1.5 + 6 = 11 kW ; Latent = 1.1 kW

Total = 12.1 kW = 12.1/3.517 = 3.44 TR ; RSHF = 11/12.1 = 0.91

12. Solved Numerical Problems

Problem 1 : Bell–Coleman cycle

Air at 1 bar, 10 °C is compressed to 5 bar, cooled to 30 °C and expanded to 1 bar. Find COP and mass flow per TR ($c_p = 1.005$, $\gamma = 1.4$).

$r_p^{0.2857} = 5^{0.2857} = 1.584$; $T_2 = 283 \times 1.584 = 448$ K ; $T_4 = 303/1.584 = 191$ K

RE = 1.005(283 – 191) = 92.2 kJ/kg ; W = 1.005[(448 – 283) – (303 – 191)] = 53.8 kJ/kg

COP = 92.2/53.8 = 1.71 = 1/(1.584 – 1) ; $\dot{m}$ per TR = 3.517/92.2 = 0.038 kg/s

Problem 2 : VCR cycle (enthalpies from p-h chart)

$h_1 = 250$, $h_2 = 290$, $h_3 = h_4 = 95$ kJ/kg. Find RE, work, COP, flow rate and power per TR.

RE = 250 – 95 = 155 kJ/kg ; W = 290 – 250 = 40 kJ/kg ; COP = 155/40 = 3.875

$\dot{m}$ = 3.517/155 = 0.0227 kg/s ; Power = 0.0227 × 40 = 0.91 kW per TR

Problem 3 : Carnot COP for comparison

Evaporator −10 °C (263 K), condenser 35 °C (308 K): $COP_{Carnot} = 263/45 = 5.84$

Problem 4 : Fresh air and recirculated air mixing

See §7 example: $t_3 = 26.7$ °C, $\omega_3 = 0.016$ kg/kg.

13. Quick Revision & Important Questions

- $COP_R = Q_2/W$; 1 TR = 3.517 kW; Bell–Coleman: $COP = 1/(r_p^{(\gamma-1)/\gamma} - 1)$.
- VCR: $RE = h_1 - h_4$, $W = h_2 - h_1$, $h_3 = h_4$.
- $\omega = 0.622\, p_v/(P - p_v)$; $SHF = Q_s/(Q_s + Q_l)$; ET combines DBT, humidity, velocity.

→ Explain the Bell–Coleman cycle with T-s diagram. Derive its COP.

→ Explain the working of a vapour-compression refrigeration system with p-h and T-s diagrams.

→ List desirable properties of a refrigerant. What do R-12, R-22, R-134a, R-717 mean?

→ Define DBT, WBT, DPT, RH, humidity ratio. Describe the psychrometric processes.

→ Explain effective temperature and the comfort chart. How is cooling load

estimated?

~ End of Unit V ~